

Sales Profitability Analysis

Investigating Consumer Segment Profitability in the Superstore Dataset

Business Analysis Report

1. Business Problem

The Consumer segment experiences lower profitability, with Bookcases representing a significant area of concern.

The analysis suggests that an unfavorable product sales mix contributes to this issue. Consumer sales are more heavily weighted toward lower-margin products, while higher-margin products contribute insufficient profit to offset losses.

Higher discount levels are also associated with declining profit margins, potentially amplifying the profitability problem.

These findings highlight the need to review product mix and discount strategies to improve profitability without unnecessarily sacrificing sales.

2. Analysis Approach

The analysis followed a hypothesis-driven approach to identify the factors contributing to lower profitability in the Consumer segment.

Step 1 — Profitability Assessment

Compared sales, profit, and profit margins across customer segments, product categories, and sub-categories. This identified Furniture as a low-margin category, with Bookcases and Tables requiring further investigation.

Step 2 — Discount Analysis

Examined profitability across discount levels and customer segments. High-discount transactions were associated with significant losses in both Consumer and Corporate, suggesting that discounts alone might not fully explain the profitability gap.

Step 3 — Within-Product Comparison

Compared the profitability of the same Bookcases products across Consumer and Corporate under discounts below 30% to investigate whether differences remained when comparing identical products.

Step 4 — Product Mix Analysis

Applied shared product-level profit margins to each segment's actual sales distribution, isolating the contribution of product mix under a common-margin assumption.

The resulting expected margins were 6.22% for Consumer and 14.29% for Corporate, supporting the hypothesis that differences in product sales mix contribute to Consumer's lower profitability.

3. Key Findings

Finding 1 — Consumer Has the Lowest Overall Profit Margin

Consumer generates substantial sales but has the lowest overall profit margin among the three customer segments. Furniture is a particularly weak-performing category.

Finding 2 — Bookcases and Tables Have Distinct Profitability Problems

Tables generates negative profit margins across all three customer segments, indicating a broader profitability issue.

Bookcases performs particularly poorly in Consumer, while Corporate and Home Office show positive margins.

Finding 3 — Higher Discounts Are Associated with Lower Profitability

Higher discount levels are associated with declining profit margins and greater losses. However, the available data does not establish whether these discounts generate sufficient additional sales to justify their cost.

Finding 4 — Product Mix Contributes to the Consumer Profitability Gap

Using shared product-level margins for Bookcases transactions with discounts below 30%, the expected profit margins were:

- Consumer: 6.22%
- Corporate: 14.29%

The 8.07-percentage-point difference suggests that Consumer's product sales mix contributes to its weaker profitability, independently of the product-level margin assumptions used in this comparison.

4. Business Recommendations

Recommendation 1 — Evaluate Discount Effectiveness

Investigate whether higher discounts generate enough incremental sales and contribution profit to offset reduced margins.

Compare alternative pricing and discount strategies using controlled tests where feasible, with total profit as the primary success metric.

Recommendation 2 — Improve Consumer Bookcases Product Mix

Identify profitable Bookcases products with growth potential and investigate opportunities to increase their sales contribution through merchandising, targeted promotions, and product recommendations.

Recommendation 3 — Review Tables Profitability

Investigate pricing, product costs, and discount practices across all customer segments to identify opportunities for improving Tables profitability.

5. Limitations and Next Steps

The analysis has several limitations:

- **Causality:** Historical discount and sales data cannot establish whether discounts directly increase sales volume.
- **Missing business context:** Information on promotion objectives, inventory, customer

acquisition costs, and returns is unavailable.

- **Product Mix:** The standardized-margin comparison supports a product mix explanation but does not establish that product mix is the only contributor to the profitability gap.

Recommended Next Steps

1. Obtain additional historical promotion and pricing data.
2. Evaluate discount effectiveness through controlled experiments where feasible.
3. Identify higher-margin Bookcases products with potential for increased demand.
4. Monitor total profit, profit margin, sales volume, and product mix after implementing changes.